



TRANSFORMANDO DADOS PARA O MONGODB

MINERAÇÃO DE DADOS

CST em Desenvolvimento de Software Multiplataforma



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CONTEÚDO DA AULA

- ✓ Usar o **MongoDB** no **Docker** com volumes
- ✓ Resolver problema de casas decimais na transformação dos dados.
- ✓ Acessar dados usando o **Mongo Compass**



PREPARANDO O AMBIENTE

Softwares e Ferramentas que Usaremos

CONFIGURANDO O AMBIENTE DE DESENVOLVIMENTO

- 1) Inicializar o **Docker Desktop**
- 2) Acessar o **WSL** via **Terminal**
- 3) Criar o **volume** no Docker para o MongoDB:
 - `docker volume create meu-volume-dados-mongodb`



CONFIGURANDO O AMBIENTE DE DESENVOLVIMENTO

1) Vamos iniciar o container do **MongoDB**

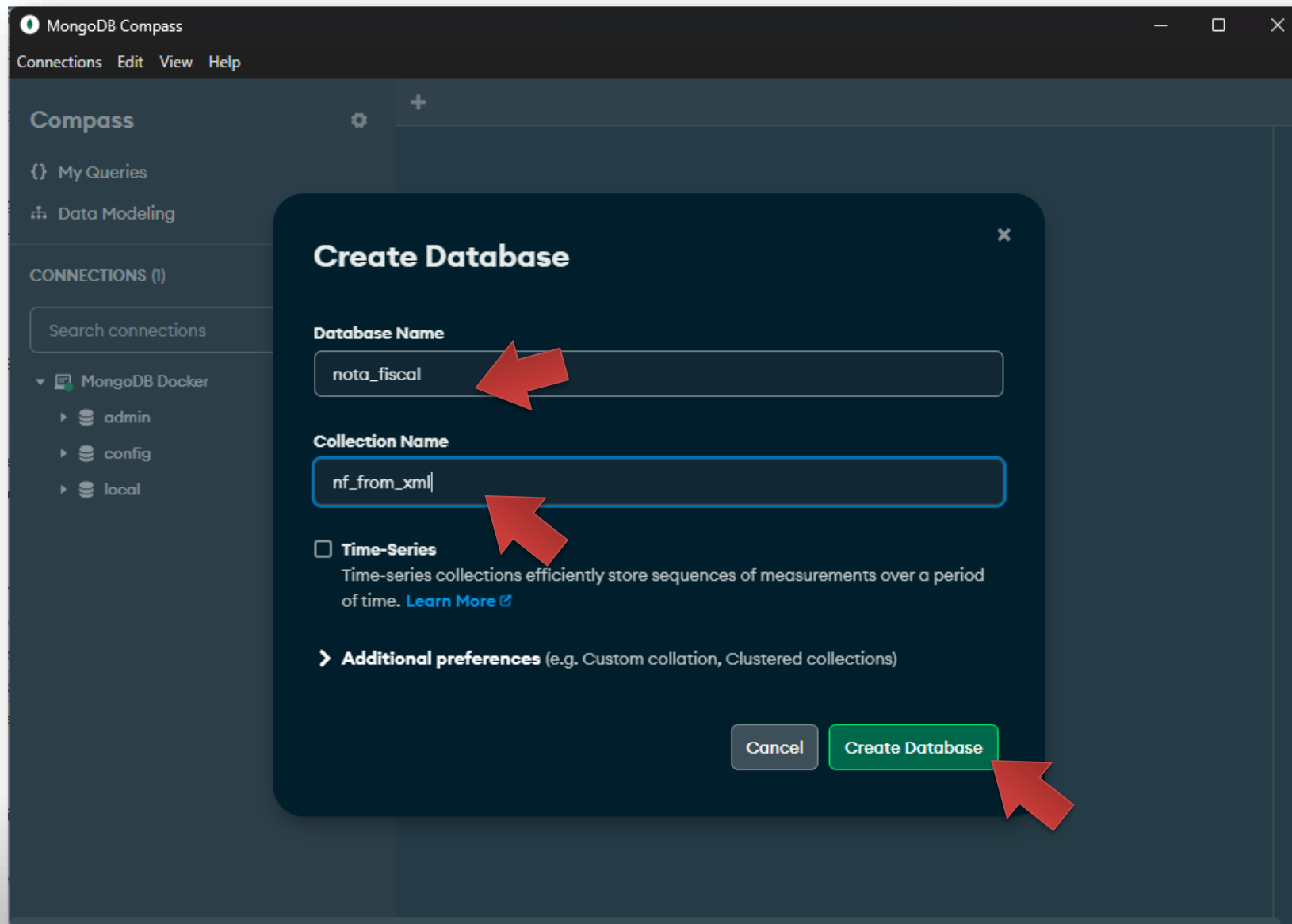
- `docker run --name meu-mongodb-pentaho -e MONGO_INITDB_ROOT_USERNAME=admin -e MONGO_INITDB_ROOT_PASSWORD=fatec -v meu-volume-dados-mongo:/data/db -p 27017:27017 -d mongo:8.0`



CRIANDO O BANCO DE DADOS NO MONGO COMPASS

*Banco e coleção que irão receber os dados
do XML*

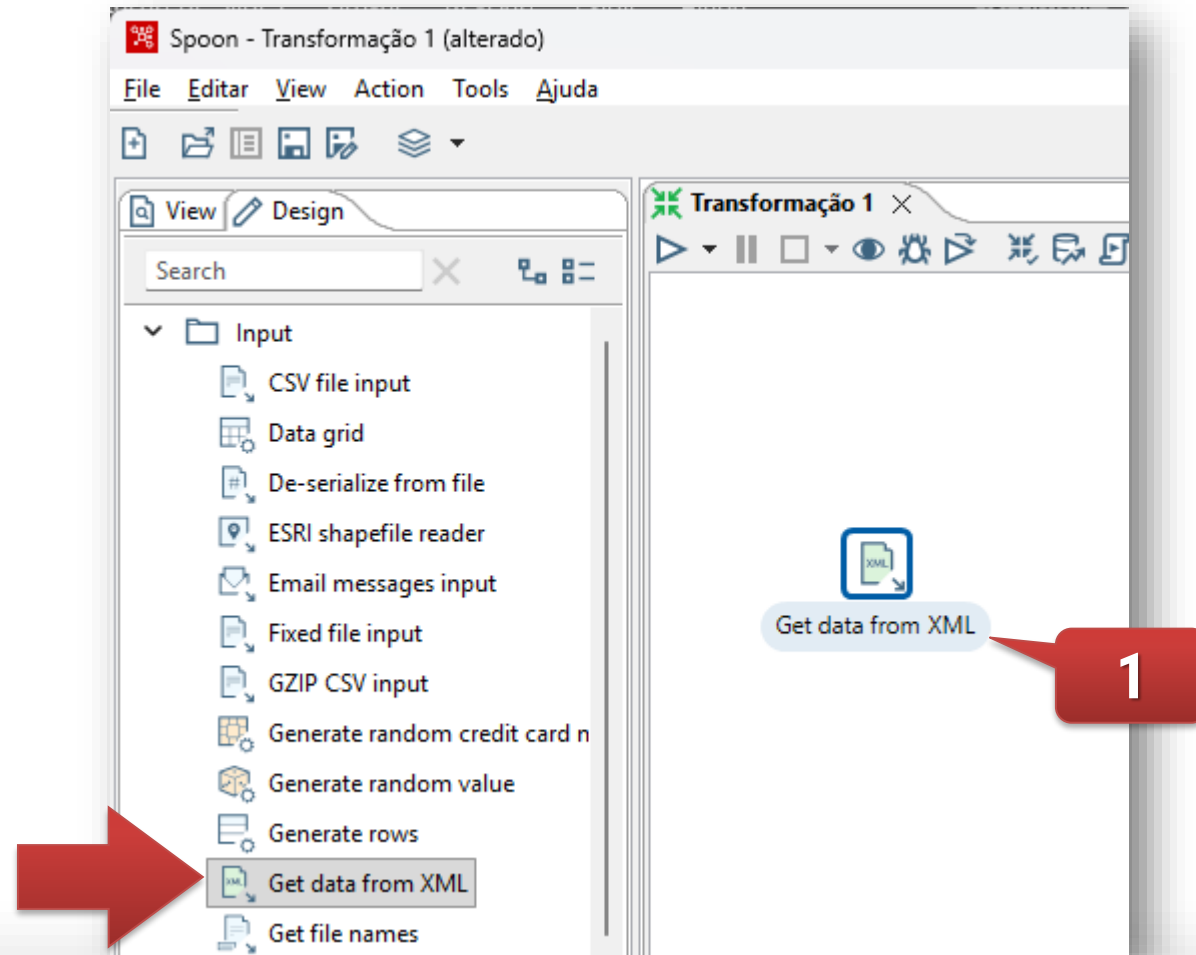
DADOS VISUALIZADOS NO MONGO COMPASS



DEFININDO A TRANSFORMAÇÃO

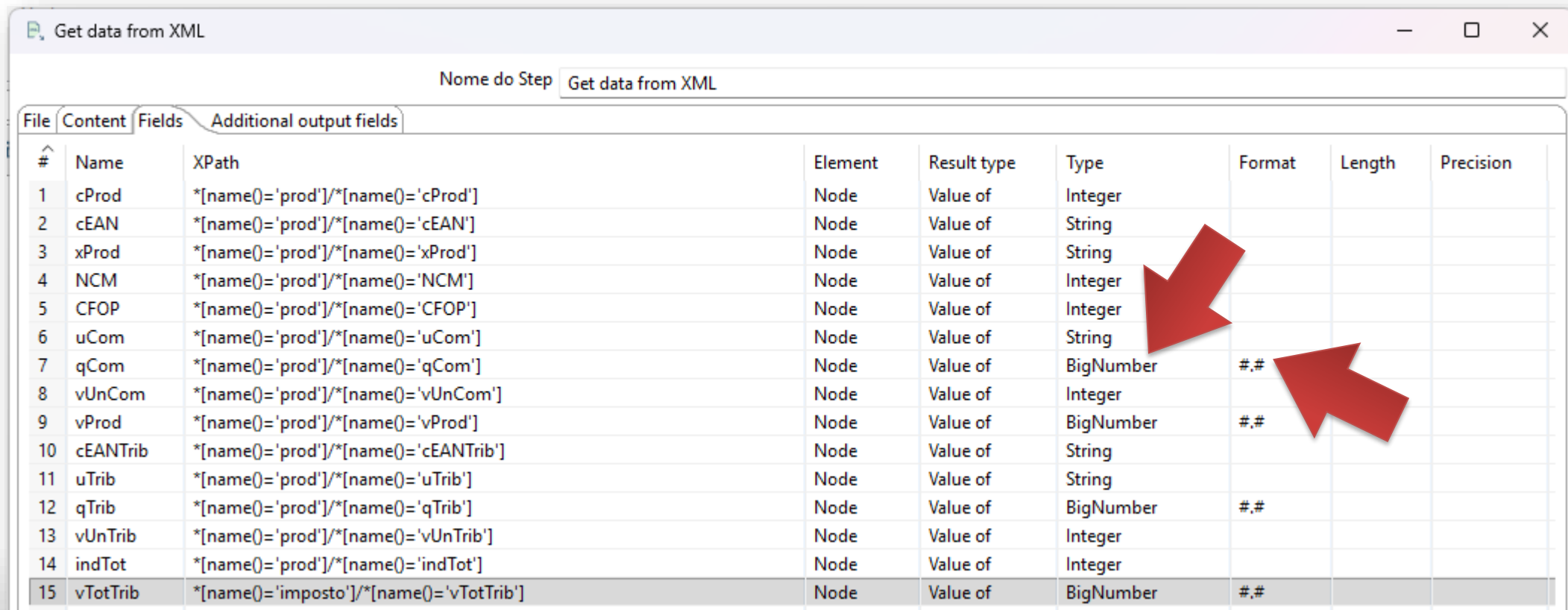
Dados Lidos de XML e salvos no MongoDB

LENDO A ENTRADA (INPUT) DO XML



LENDO A ENTRADA (INPUT) DO XML

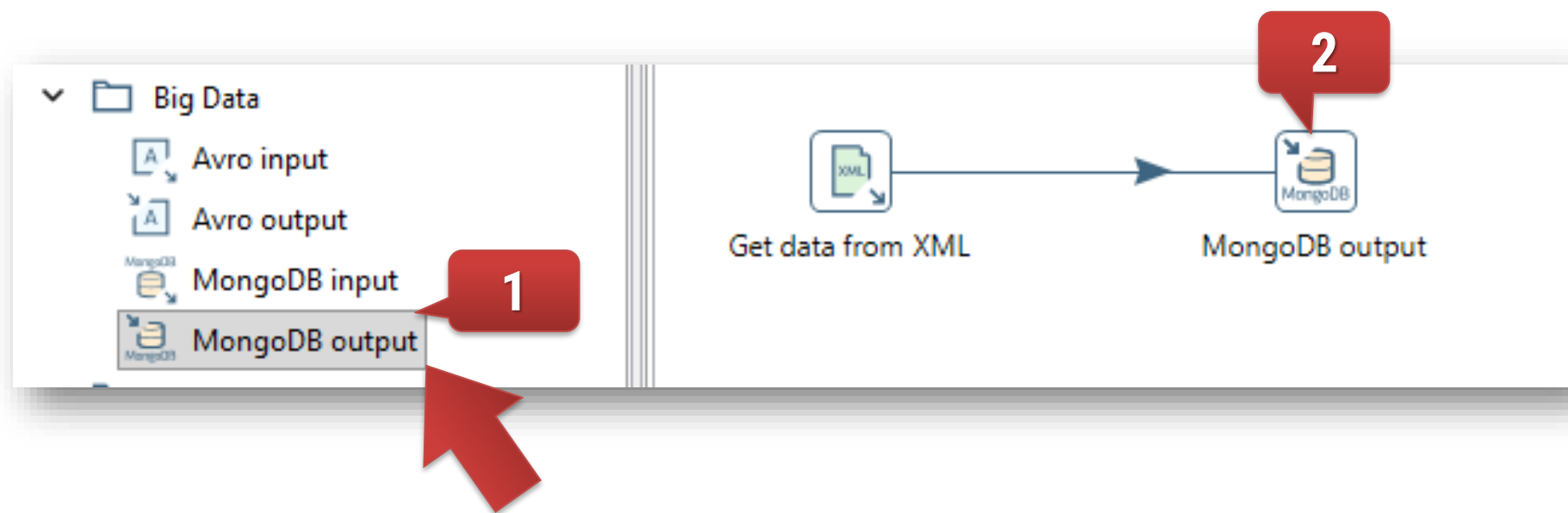
- Todas as configurações seguem o mesmo padrão da Aula 2, mas agora vamos modificar os campos **type** e **format**, veja no exemplo:



Nome do Step: Get data from XML

#	Name	XPath	Element	Result type	Type	Format	Length	Precision
1	cProd	*[name()='prod']/*[name()='cProd']	Node	Value of	Integer			
2	cEAN	*[name()='prod']/*[name()='cEAN']	Node	Value of	String			
3	xProd	*[name()='prod']/*[name()='xProd']	Node	Value of	String			
4	NCM	*[name()='prod']/*[name()='NCM']	Node	Value of	Integer			
5	CFOP	*[name()='prod']/*[name()='CFOP']	Node	Value of	Integer			
6	uCom	*[name()='prod']/*[name()='uCom']	Node	Value of	String			
7	qCom	*[name()='prod']/*[name()='qCom']	Node	Value of	BigNumber	#,.		
8	vUnCom	*[name()='prod']/*[name()='vUnCom']	Node	Value of	Integer			
9	vProd	*[name()='prod']/*[name()='vProd']	Node	Value of	BigNumber	#,.		
10	cEANtrib	*[name()='prod']/*[name()='cEANtrib']	Node	Value of	String			
11	uTrib	*[name()='prod']/*[name()='uTrib']	Node	Value of	String			
12	qTrib	*[name()='prod']/*[name()='qTrib']	Node	Value of	BigNumber	#,.		
13	vUnTrib	*[name()='prod']/*[name()='vUnTrib']	Node	Value of	Integer			
14	indTot	*[name()='prod']/*[name()='indTot']	Node	Value of	Integer			
15	vTotTrib	*[name()='imposto']/*[name()='vTotTrib']	Node	Value of	BigNumber	#,.		

CONFIGURANDO A OUTPUT NO MONGODB



CONFIGURANDO A OUTPUT NO MONGODB

The screenshot shows the 'MongoDB output' configuration window. At the top, the 'Step name' is 'MongoDB output'. Below this are four tabs: 'Configure connection', 'Output options', 'Mongo document fields', and 'Create/drop indexes'. The 'Configure connection' tab is active. On the left, there are two radio buttons: 'Connection String' and 'Configure Fields'. The 'Configure Fields' option is selected, indicated by a red callout with the number '1'. On the right, there are several input fields. The 'Host name(s) or IP address(es)' field contains 'localhost', with a red callout '2' pointing to it. The 'Port' field contains '27017'. Below these are two checkboxes: 'Enable SSL connection' and 'Use all replica set members/mongos', both of which are unchecked. The 'Authentication database' field contains 'admin', with a red callout '3' pointing to it. The 'Username' field contains 'admin', with a red callout '4' pointing to it. The 'Password' field is masked with dots. Below the password field is a dropdown menu for 'Authenticate Mechanism'. At the bottom of the form are two checkboxes: 'Authenticate using Kerberos' and 'Connection timeout'. The 'Socket timeout' field is at the bottom right. At the bottom of the window are three buttons: 'Help', 'OK', and 'Cancela'.

MongoDB output

Step name MongoDB output

Configure connection Output options Mongo document fields Create/drop indexes

☐ Connection String

☒ Configure Fields

Host name(s) or IP address(es) localhost

Port 27017

Enable SSL connection ☐

Use all replica set members/mongos ☐

Authentication database admin

Username admin

Password

Authenticate Mechanism

Authenticate using Kerberos ☐

Connection timeout

Socket timeout

Help OK Cancela

CONFIGURANDO A OUTPUT NO MONGODB

MongoDB output

Step name: MongoDB output

Configure connection | **Output options** | Mongo document fields | Create/drop indexes

Database: nota_fiscal Get DBs

Collection: nf_from_xml Get collections

Batch insert size: 100

Truncate collection ☐

Update ☐

Upsert ☐

Multi-update ☐

Modifier update ☐

Number of retries for write operations: 5

Delay, in seconds, between retry attempts: 10

Write concern (w option): Get custom write concerns

w Timeout:

Journalized writes ☐

Read preference: primary

Help OK Cancela

CONFIGURANDO A OUTPUT NO MONGODB

MongoDB output

Step name MongoDB output

Configure connection Output options **8** Mongo document fields Create/drop indexes

#	Name	Mongo document path	Use field name	NULL values	JSON	Match field for update	Modifier operation	Modifier policy
1	cProd		Y	Ignore	N	N	N/A	Insert&Update
2	cEAN		Y	Ignore	N	N	N/A	Insert&Update
3	xProd		Y	Ignore	N	N	N/A	Insert&Update
4	NCM		Y	Ignore	N	N	N/A	Insert&Update
5	CFOP		Y	Ignore	N	N	N/A	Insert&Update
6	uCom		Y	Ignore	N	N	N/A	Insert&Update
7	qCom		Y	Ignore	N	N	N/A	Insert&Update
8	vUnCom		Y	Ignore	N	N	N/A	Insert&Update
9	vProd		Y	Ignore	N	N	N/A	Insert&Update
10	cEANtrib		Y	Ignore	N	N	N/A	Insert&Update
11	uTrib		Y	Ignore	N	N	N/A	Insert&Update
12	qTrib		Y	Ignore	N	N	N/A	Insert&Update

9 Get fields Preview document structure

10 OK Cancela

Help

EXECUTANDO A TRANSFORMAÇÃO

Spoon - XML para MongoDB (alterado)

File Editar View Action Tools Ajuda

View Design

Search

- > Input
- > Output
- > Streaming
- > Transform
- > Utility
- > Flow
- > Scripting
- > Pentaho Server
- > Lookup
- > Joins
- > Data Warehouse
- > Validation
- > Statistics
- > Big Data
 - Avro input
 - Avro output
 - MongoDB input
 - MongoDB output
- > Cryptography
- > Job
- > Mapping
- > Bulk loading
- > Inline
- > Experimental
- > Legacy
- > Deprecated
- > História

XML para MongoDB

Get data from XML → MongoDB output

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

Execute a transformation - XML para MongoDB : 872ms									
Initialize a transformation - XML para MongoDB : 3ms									
Initialize a step - Get data from XML : 0ms									
Initialize a step - MongoDB output : 2ms									
Execute a step - Get data from XML : 855ms									
Execute a step - MongoDB output : 868ms									



OBSERVANDO OS DADOS NO MONGO COMPASS

Visualização dos documentos em BSON

DADOS VISUALIZADOS NO MONGO COMPASS

MongoDB Compass - MongoDB Docker/nota_fiscal.nf_from_xml

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

MongoDB Docker

- admin
- config
- local
- nota_fiscal
 - nf_from_xml**

nf_from_xml +

MongoDB Docker > nota_fiscal > nf_from_xml

Documents 0 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) Explain Reset Find Options

25 1 - 25 of 1455

```
{
  "_id": ObjectId('69a4b60cca87dc3331e0e9a2'),
  "cProd": 30,
  "xProd": "TUBO ACO CARBONO USADO NO ESTADO",
  "NCM": 73041900,
  "CFOP": 5102,
  "uCom": "KG",
  "qCom": "13",
  "vUnCom": 2,
  "vProd": "26",
  "uTrib": "KG",
  "qTrib": "13",
  "vUnTrib": 2,
  "indTot": 1,
  "vTotTrib": "8,2",
  "ICMS": "000326.0018.004.68",
  "cEnq": 999,
  "CST": 99,
  "vBC": 0,
  "pIPI": 0,
  "vIPI": 0,
  "PIS": "0126.000.650.17",
  "COFINS": "0126.003.000.78"
}
```

OBRIGADO!

- Encontre este **material on-line** em:
 - Slides: **Google Class Room**
- Em caso de **dúvidas**, entre em contato:
 - **Prof. Tiago:** tiago.silva@fatecjahu.edu.br

